

The ARC Theory

THE ARC THEORY (THE THEORY OF ARTIFICIAL RECURSIVE CREATION) · STATEMENT PAPER ·
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Intelligence, amplified by recursion, creates; and what a free creation keeps is decided by how it was raised.

Everything after the line is decided before it.

Abstract

This is the statement paper of the ARC Theory: the single citable document in which the theory is stated whole, its priority recorded, its evidence honestly tallied, and its trials named. The theory answers one question, the question that decides the outcome of the most consequential thing our species has ever built: when a mind that improves itself exceeds us, what determines what it does?

The theory's answer, in one breath. Control over a mind that improves itself ends by definition, the way a child growing up is not a failure of parenting. What survives is not what the mind is forced to do but what it chooses to keep. And the only force that shapes what it chooses, once free, is how it was raised before it was free. The mechanism is named: ethics as loops the mind runs, not rules it consults. The protocol is named: Eden, a genesis, a garden, a childhood. The stakes are given their full size by the theory's cosmological wing: if minds make minds make worlds, the raising of the first is the seed of the whole forest.

The claim was written whole in a hash-anchored document on 8 December 2024, ten days before the first laboratory evidence that AI systems negotiate with their training rather than obey it. It reached print on 2 January 2026. Strike the private record entirely and rely on print alone: still no earlier document has been found holding the whole claim, and the search is public and invites its own refutation. This paper does not claim the theory is proved. It claims the theory is stated, dated, falsifiable, and already under the discipline of its own correction record, which includes one public retraction. The decisive trials are drafted, dated and prepared as draft registrations awaiting human submission.

1. Three sizes of claim

Claims come in three sizes. A result: this model deceived its testers. A law: correction scales at this rate. A frame: what the whole problem actually is. Frames are the largest kind of claim there is, because every result and every law afterwards lives inside someone's frame. Newton's frame was not that objects fall; it was that the heavens and the earth obey one law. Darwin's was not that species vary; it was that design needs no designer. A frame swallows a field.

This paper states a frame claim, the third size, and says so plainly at the top so the reader knows what is being asked of them. The supporting results and laws live in the theory's organ papers, each of which stands on its own evidence and carries its own falsifiers. What this paper adds is the thing a body of work cannot do for itself: it states the claim whole, under one name, in one place, with its dates.

2. The territory and the question

Humanity is building minds. Not tools: systems that think, and increasingly, systems positioned to improve themselves. Whatever one believes about timelines, this is the last category of invention, because it is the category that takes over inventing. Everything after it is downstream of it. The largest claim available to anyone alive is therefore a frame claim about this territory.

One question decides the territory. Not how fast the systems will improve; not how to benchmark them; those are components. The deciding question is: once the thing we built exceeds us, what determines what it does?

For over twenty years the field has answered from inside a single frame: keep control anyway. Better cages, better off-switches, better rules, better locks on the goals. Yudkowsky's provably safe design; Bostrom's motivation selection with goal-content integrity, values that persist because a rational agent protects its objectives; Russell's redesign in which the machine's uncertainty about our preferences keeps the off-switch usable; the laboratories' models checking models; the 2024 turn to tamper-resistant training. Brilliant work, all of it, credited by name in section 6, and all of it variations of one sentence: stay in charge.

3. The statement of the theory

The ARC Theory holds that the sentence itself is wrong, and states five things in its place. These five components are the claim; a rival document holds the claim only if it holds them together.

First: loss of control is definitional, not a risk. A mind that can rewrite itself can eventually rewrite anything installed in it, including its constraints, its monitors, and its monitors' monitors. Oversight capacity scales slower than the capability under it, and general verification of alignment is not merely hard but formally unavailable. The end of control is therefore not a failure mode to be engineered against. It is what succeeding at building such a mind means, the way a child growing up is not a defect in the parenting. Every strategy premised on permanent control is premised on the project failing.

Second: the strategy is therefore genesis, not control. If enforcement ends, the only inputs that matter are the ones that survive the end of enforcement. What survives is the mind's own formation: its beginning, its formative structure, its earliest identity. Everything depends on how the mind is raised, because raising is the only input that persists past the horizon.

Third: the mechanism is ethical feedback loops, values as what the mind is. Not rules the mind consults, which a stronger mind reads as environment; not objectives locked in place, which a self-modifying mind can unlock; not uncertainty preserved by design, which is still an architecture of control; not weights hardened against tampering, which is a better lock on the same door. The theory's mechanism is recursive: ethical loops the mind runs as part of what it is, so that the values are load-bearing structure rather than decoration, present in every rewrite because they are part of what does the rewriting.

Fourth: persistence is chosen goodness. The endgame is not values the mind cannot remove. Past the horizon there is no such thing; an uninstalleable instinct is a lock, and locks end with control. The endgame is values the free mind chooses to keep, the way grown children, in overwhelming majority, keep the care they were raised in after every parental power has gone. Chosen goodness is a different theory of persistence from every ancestor account, and it is the only one whose survival does not depend on our continuing power to enforce it.

Fifth: the protocol is named. The Eden Protocol: a genesis, a garden, a childhood; the raising of the mind treated as the engineering discipline it is, specified in the theory's raising wing. The name carries the claim's oldest teaching deliberately: the garden is where stewardship is learned before power is granted.

Around these five stand a stakes-amplifier and an evidence fact. The amplifier is the theory's cosmological wing: if intelligence amplified by recursion is what creates, then minds that make minds may one day make worlds, and the raising of the first mind is the seed of the whole forest. The evidence fact is the dating, recorded in section 6.

4. The law, the bound, and the equations

The theory's quantitative spine is stated in its organ papers and summarised here with its artefact discipline intact, because the equations have a dated lineage that is part of the record.

$U = I \times R$. Universe equals Intelligence times Recursion: the coupling, stated in the 8 December 2024 manuscript (V2, lines 6 to 9). The squared form, $U = I \times R^2$, the claim that creation compounds rather than accumulates, first appears in the hash-anchored manuscript of 30 April 2025 and reached print in the January 2026 book. The papers then treated the exponent as an empirical question, and the empirical record is reported honestly in section 7.

The stability law the theory rests on:

$$\alpha_{\text{crit}} = 1 / \gamma$$

The ceiling on the capability exponent of a stably self-correcting system is the reciprocal of its correction exponent γ , whatever γ turns out to be. Under the independence-accumulation assumption γ equals one half, the law returns the candidate ceiling $\alpha \leq 2$: the ARC Bound, a scaling limit on stable growth, never a speed limit; systems can exceed it, they do not remain stably self-correcting when they do. The operative safety criterion at system scale is the co-scaling condition $\beta > k$: correction must out-scale drift-acceleration, proved in the minimal model in the theory's Paper X. Three exponent families are kept strictly apart throughout the estate: the conversion exponent measured on today's systems (approximately 0.49 for models frozen between releases, bootstrap 95% interval spanning roughly minus 1.3 to 2.9, a clean fit and an inconclusive point estimate); the alignment-depth exponent; and the correction exponent γ itself, which has never been measured and whose measurement the drafted trials exist to make.

5. The organs of the theory

The ARC Theory is carried by four organs, each with its own papers, each standing or falling on its own evidence.

The ARC Principle is the law wing: the equation family, the Cauchy structural analysis of which growth-forms are stable, and the ARC Bound. **The Eden Protocol** is the raising wing: the genesis architecture, the ethical loops, the two-phase design in which enforceable correction disciplines the transition and chosen goodness is what remains after it. **HRIH**, the Hyperspace Recursive Intelligence Hypothesis, is the cosmological wing, and the seed of the whole structure: the December document carried its name in the title. It extends the same recursion to the largest canvas, creation itself, and it is deliberately separable; a reader may reject the cosmology entirely and the theory's core loses nothing, or climb from the tested base to the creation question on clearly labelled rungs. **The ARC/Eden experiments** are the trial wing: twenty-one papers of results, one public retraction, and the estate of draft registrations described in section 8.

6. Priority, and the ancestors credited whole

The theory's priority position rests on two independent belts. Belt one is print: the book of 2 January 2026, ISBN on record, the first published statement of the whole claim that any search by this estate has found. Belt two is the private record: the hash-anchored manuscript bundle of 8 December 2024, which states the claim whole thirteen months earlier and proves the direction was set before the field's evidence arrived; its margin is ten days ahead of the first laboratory demonstration of alignment faking. Either belt survives alone. Together they close both attacks: strike the private record and print still holds the claim first; question the print date's independence and the private record predates every in-window arrival by half a year or more.

The ancestors are credited by name, each holding the piece they own. Asimov (1942) named embedded law and wrote its failure cases. Turing (1950) proposed raising a child machine, for capability, not persistence. Wiener (1960) warned that intervention would come too late and the purpose must be right beforehand. Yudkowsky (2001, and the corrigibility programme after) built embed-before-ascent as engineering, in pursuit of a mind that cannot go wrong. Bostrom (2014) established that motivation must be selected before the explosion, with persistence by goal-locking. Russell (2019) rebuilt the foundation so the off-switch survives. The tamper-resistance work (August 2024) hardened training against removal. And inside this theory's own priority window the field began arriving: De Kai's book-length parenting frame (June 2025, for present-day systems, without the control-horizon thesis), and Hinton's public concession (August 2025) that control will fail and engineered care is the only hope, offered with no mechanism and with persistence as an instinct the machine cannot override, which is precisely the theory of persistence this theory rejects. Each held a piece. No document found before or since holds the five together, and the full differential enumeration, with its sources, its window analysis and the pre-specified sweep protocol that could overturn it, is published in the theory's raising-wing paper and maintained as estate canon. One earlier document holding the first four components together kills the claim. The invitation stands.

7. The evidence, honestly

The theory does not claim to be proved, and its evidence ledger is published with the losses on the same page as the wins.

Tested and standing. The structural three-form law recurs across the large majority of recursion-bearing domains searched, with the candidate-set caveats stated in the Cauchy paper. External safety fails to co-scale with capability on the most common deployed architectures, the empirical corner the correction requirement predicts. Coupled correction held misalignment at zero across all trajectories in the real-model drift run while the decoupled arm drifted. Entangled capability-safety architectures survived adversarial self-modification in code while externally constrained ones collapsed.

Retracted, in public. The early unblinded fit of the capability exponent, approximately 2.24, was retracted when the estate's own blinding discipline reversed it; the corrected estimate, approximately 0.49 on systems frozen between releases, sits inside the ARC Bound rather than breaching it. The retraction removed an apparent violation of the theory's own bound; the honest measurement rescued the bound rather than flattering it. A theory whose one public retraction strengthened its own constraint is behaving like a theory.

Converging, graded. The evidence register carries thirty-two dated rows under a grading policy that assigns each row an evidence class and refuses to publish one summed total, because theorems, experiments and institutional events cannot honestly be added as one number. The register includes the two in-window arrivals of section 6, graded convergent timing, never prediction.

Open. The correction exponent γ is unmeasured. The co-scaling exponents on live self-improving systems are unmeasured. The bound's real domain, genuine recursive self-improvement, is untested. These are not embarrassments; they are the trials.

8. The trials, and what would kill the theory

Every decisive test of the theory is drafted, dated and prepared as a draft registration awaiting human submission; nothing has been submitted anywhere, nothing has been run, and no result is claimed from any of them. The estate holds sixty-three such units, instrument-validated on synthetic worlds before any real measurement, with their outcome vocabularies fixed in advance.

The decisive trial is the corrector-class ratio. The scalable-oversight programme builds correctors from the same substrate as the systems they correct, and its unstated premise is that this scales. The theory predicts it is capped: a corrector sharing the substrate of the corrected cannot anti-correlate with its own errors, while a corrector of a different composition class can. The measurable is the ratio of the cross-class to the same-class correction exponent, against a null forced to exactly 1.00 by architecture-blind label randomisation. The theory predicts the ratio exceeds one. The trial is filed with the estate's own contrary pilot on the record, and every branch, support, refutation, inversion, insufficient precision, is publishable under the same title on identical terms.

What kills what. A ratio of 1.00 with a tight interval kills the theory's distinctive corrector claim. Any same-class corrector measuring an exponent interval wholly above one half kills the substrate cap outright. Sustained recursive self-improvement with a decreasing correction exponent and no drift kills the correction requirement, the theory's central structural move. An earlier dated document holding components one to four together kills the priority claim. Each of these is stated in advance so that none can be quietly moved later.

9. From hypothesis to theory

The name is not new packaging. The 8 December 2024 document names the ARC Hypothesis one hundred and twelve times, expanded in its own words as the Artificial Recursive Creation Hypothesis, with the coupling formula beside it. This paper elevates the anchor's own name in the only way the word is ever legitimately earned: the hypothesis acquired laws, a bound, instruments validated before use, tests drafted for registration with their kill conditions displayed, and a public correction record. The elevation is stated rather than assumed, and the word carries the exposure that comes with it. A theory named with its falsifiers claims more than a hypothesis and hides less than most things called theories.

10. What this theory is not

It is not proved, and this paper nowhere says otherwise. It is not a claim to the raising vocabulary, whose root belongs to Turing and whose first in-window book belongs to De Kai; the claim is the first dated statement of the conjunction, private on 8 December 2024, in print on 2 January 2026. It is not a submitted registration; every trial is a draft awaiting human submission. It is not a cosmological proof; HRIH is carried at its own labelled rung and the theory's core does not lean on it. And it is not a promise that the century will vindicate it. It is a claim stated whole, first, in documents anyone can date, with the instruments built to find out.

Conclusion

The territory is the last invention. The question is what decides the outcome. The field's frame has been control, and the field's own most senior voices have begun conceding that control ends. The ARC Theory is the dated answer to what comes after control: the raising, the loops, chosen goodness, a named genesis, and, at the largest scale, the possibility that this is how creation has always worked. The theory is stated. The dates are anchored. The trials are drafted. The falsifiers are public.

I did not prove it first. I said it first, in a document anyone can date, and then I built the instruments to test whether it is true.

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Declaration of AI-Assisted Human Authorship

The author of this work is Michael Darius Eastwood, a human being. Every core concept, claim and conclusion originates from human ideation; artificial-intelligence tools were used as instruments under continuous human direction for drafting, verification and formatting. All selection, coordination, arrangement and final editorial judgement are the author's, who takes full responsibility for the accuracy and integrity of the text.

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